Executable Operating Contracts for Evidence Reuse in Aircraft Localization and Inspection

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Abstract

Changes in inspection requirements and components alter the statistical evidence applicable to an engineering output. We introduce executable operating contracts for evidence reuse in aircraft localization and inspection. A typed task–source–evidence representation links consumer events, committed policies, numerical source versions, producer dependencies and calibrated cards. Event containment and invariant requested functions govern reuse, while ordered source replacement and persistent history govern execution and issuance. On the same 47 supported event contexts and fixed policies, the selected anchor catalogue reduces evidence-preparation event–record evaluations by 20%. Across 705 producer edits of initially admitted contexts, the implementation withdraws 489 affected statements and retains 216 applicable ones, with all decisions agreeing with the declared dependency oracle and all original contexts restored. Evaluation checks 15,360 deterministic implementation decisions and includes 13,824 independent synthetic localization episodes, 36 complete-route inputs and two image-task studies with 12,800 independent inputs. At risk allowance 0.15, reuse supports 24 and 23 of 160 unique image-event contexts; cached direct renewal supports 30 and 29. The framework provides traceable admission, selective withdrawal and restoration of evidence across changing engineering tasks.

Keywords: Executable operating contracts, Evidence reuse, Anytime localization, Risk calibration, Aircraft inspection

1. Introduction

Engineering changes alter the relationship between an inspection output and the evidence supporting its use. A consumer may relax an image requirement, a decoder may be replaced, or a newer localization estimate may become available after capture. Each change poses a concrete question: which existing statistical statements remain applicable, which must be withdrawn, and what computation is required to restore the requested service? Aircraft inspection provides a connected setting in which geometric estimation, image acquisition, association and reporting depend on different parts of the same information pipeline.

Anytime computation connects estimation quality to processing time [1, 2]; engineering knowledge models connect decisions to requirements and dependencies [3, 4, 5]. Risk-controlled selection attaches statistical evidence to a specified effective program [6]. An inspection service must preserve all three relationships through execution and change. Successive registration checkpoints revise one immutable scan/prior, capture consumes one installed version, and a risk card concerns a policy, population, processing service and failure event.

We introduce executable operating contracts for evidence reuse in aircraft localization and inspection (Fig. 1). A typed task–source–evidence representation compiles the consumer's requirements into executable predicates and a committed computation plan. Event-containment witnesses identify covered requirements; requested dependency projections identify applicable producer snapshots. Ordered source replacement and a persistent execution-history monitor connect these decisions to the output actually produced. The resulting interface supports evidence reuse, selective withdrawal and restoration across changing inspection requirements.

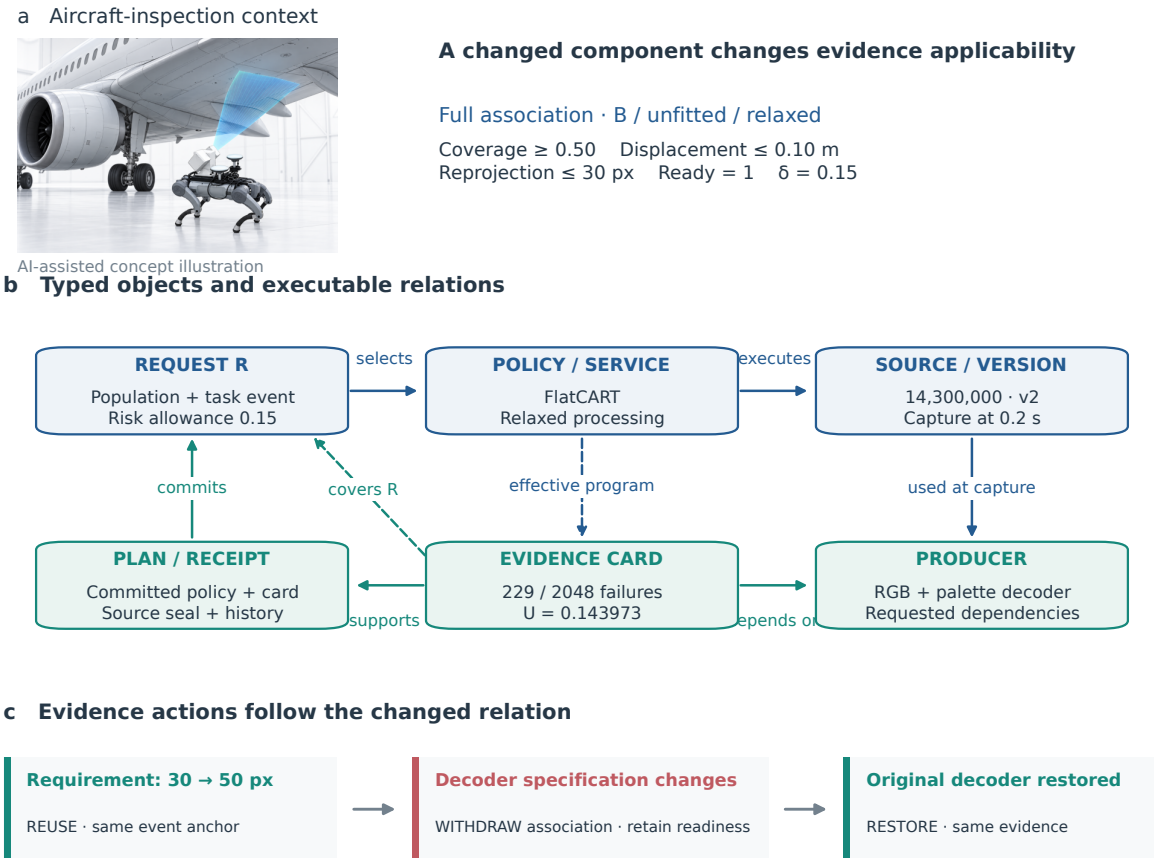


Figure 1: An executable task–source–evidence instance. (a) AI-assisted aircraft-inspection concept illustration. (b) Typed objects and named relations for the archived full-association task: the request selects FlatCART at relaxed service; capture consumes source version two; a population/event card supports the committed plan through its requested producer dependencies. (c) Requirement relaxation, decoder change and restoration produce different evidence actions. The conceptual image supplies context; quantitative evidence comes from the executed synthetic studies.

The paper makes three contributions. First, it defines an executable representation connecting task events, effective policies, numerical source versions, producer dependencies and calibrated cards. Second, it establishes the correctness conditions for the evidence lifecycle: containment preserves covered events, invariant requested functions preserve producer applicability, and persistent history prevents invalid attachments from being reissued within a source scope. Third, it evaluates these operations through independent numerical and image inputs, complete routes and a fixed change/recovery matrix. On the same 47 supported event contexts and fixed policies, anchor reuse reduces evidence-preparation event–record evaluations by 20%. Across 705 producer edits of initially admitted contexts, the implementation withdraws 489 affected statements and retains 216 applicable ones, agreeing with the declared dependency oracle throughout.

The article develops the representation and execution rules, then evaluates request support, preparation work and change decisions alongside the underlying numerical services.

2. Related work

Engineering knowledge and change operations. Process knowledge and decision knowledge graphs organize engineering choices and their supporting context [3, 4]. Semantic change propagation makes component

relations available for impact analysis [5, 7], while inspection ontologies connect operating conditions to robot requirements [8]. Their contribution is assessed through engineering operations: representational coverage and queries [9], or executable conversion and downstream model fidelity [10]. Our corresponding operation is the continued applicability of a statistical statement after a task or provider changes. It requires both the quantity's implemented dependency projection and the failure event certified by the original card.

Computation contracts and source revisions. Metareasoning and anytime algorithms allocate computation according to its expected value [11, 12, 13]. Pant et al. connect computational contracts to estimation and control [1]; Aldana-López et al. study stability-preserving perception scheduling [2]. We retain ICP and joint-state estimation [14, 15, 16], using complete-event calibration alongside approximate curvature covariance [17, 18, 19]. The interface tracks correlated checkpoints as revisions of one source, then links the version used at capture to the requested evidence. This source identity is carried through replacement and causal motion replay.

Risk selection and evidence lifetime. Assumption–guarantee contracts formalize component obligations [20]; binomial inversion and risk-controlled selection calibrate specified configurations [21, 6, 22]. We use simultaneous finite-family coverage to support before-sensing policy and anchor selection, then extend each anchor through explicit containment and requested-function invariance. Runtime assurance supplies assumption monitoring and controller switching [23]. Here the monitored action is evidence issuance, with a source-scoped history latch that persists through plan reconstruction while numerical installation continues through its original interface.

Table 1 identifies the links introduced by the operating contract relative to these established operations. The central contribution is an executable evidence lifecycle: task/event identity, producer applicability and numerical source history jointly govern admission, retention, withdrawal and restoration. The evaluation measures those operations directly through matched preparation counts and change decisions.

Table 1: Positioning by engineering operation. Each row maps an established research operation to the executable relation evaluated here; it is a conceptual comparison of scope.

Established operation	Executable relation introduced here	Evaluation endpoint
Knowledge and change reasoning [3, 4, 5]	Requested quantities link provider changes to specific certified events	Affected statements withdrawn; applicable statements retained and restored
Anytime estimation and scheduling [1, 2]	Immutable-source revisions link a committed policy to the version consumed at capture	Source trace, localization trade-offs and complete-route availability
Risk-controlled configuration selection [6, 22]	Containment and invariant requested projections link new requests to existing risk cards	Unique request support and preparation work on matched contexts
Runtime assumption monitoring [23]	Persistent source history links attachment attempts to evidence status	Invalid-packet refusal, gate reconstruction and trusted-scope recovery

3. Operating contracts for anytime localization

3.1. Typed task–source–evidence representation

The operating knowledge structure is

$$\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma), \tag{1}$$

where $\mathcal{V}$ contains entities, τ assigns types, $\mathcal{R}$ records their relations, σ stores immutable specifications and fingerprints, and Γ contains executable predicates. A request $R = (g, a, \mathcal{E}, \delta)$ fixes the population g, processing service a, required failure events $\mathcal{E}$ and allowance δ before sensing. Source revisions use an ordered

Table 2: Operating knowledge and executable decisions. Identifiers link task requirements to numerical versions and their supporting evidence.

Object	Immutable specification	Executable role
Request	Population, service, event intervals, allowance	Compile task predicates and required dependencies
Policy/service	Predictor, action rule, budget, load, deadline	Reserve work; choose continuation, publication and stopping
Source/version	Scan, prior, sequence, payload, rollback	Replace the source contribution and replay acquired motion
Producer	Quantity functions and dependency snapshot	Check the requested projection after a component edit
Evidence card	Event, producer, population, manifest, n, k, U	Verify and admit a compatible risk statement
Plan/receipt	Selected policy, card and source/version seals	Commit before sensing; issue, withdraw and restore evidence

transaction; dependency links connect each requested quantity to its implemented providers. The reference implementation represents these objects through typed records, identifier-linked relations and monitored state.

Relations such as *derived-from*, *revises*, *depends-on* and *supports* connect these roles. Figure 1b instantiates the representation: the B/unfitted full-association request selects FlatCART at relaxed service; source 14,300,000 supplies the installed capture-time version; and its applicable card records $229/2,048$ calibration failures with $U = 0.143973$. Their predicates determine a consequential action: a new source version replaces its predecessor; a changed requested dependency revokes the affected card; a covered requirement retains its bound. Every decision therefore has an inspectable task, computation and evidence path.

3.2. Immutable sources and retained computation

The robot localizes against a stationary airframe prior. The joint state $x = (t, R, b, m_t, m_R)$ contains robot translation and rotation, six motion-bias components and map translation and rotation, with an 18-dimensional local covariance P. Its airframe-relative position error is

$$e_p = \widehat{m}_R^\top(\widehat{t} - \widehat{m}_t) - (m_R^\star)^\top(t^\star - m_t^\star). \tag{2}$$

Measured motion propagates the joint state and covariance using an $SE(3)$ linearization [16]. Simulator truth supplies offline labels; policy decisions use causal estimates and declared specifications.

A source s consists of a scan and its immutable prior (x_s^-, P_s^-). Robust point-to-plane registration minimizes

$$\Phi_s(x) = d^\top (P_s^-)^{-1} d + \sigma^{-2} \left[\sum_{i \in \mathcal{I}(x)} \rho(r_i(x)) + (n - |\mathcal{I}(x)|)\rho(r_{\text{out}}) \right], \quad d = x \boxminus x_s^-. \tag{3}$$

The unmatched-point penalty keeps difficult correspondences in the objective. We use at most 96 deterministically thinned points, 25 iterations, Huber scale 0.0625 m, $\sigma = 0.025$ m and $r_{\text{out}} = 0.8$ m. Checkpoints occur at iterations 1, 2, 4, 8, 16 and 25 or termination. An eligible candidate has a descending step, at least 12 pairs, clipped residual RMS below 0.065 m and more than 60% of residuals within 0.10 m. Final acceptance also requires translation and rotation steps below 0.001 m and 0.0005 rad. Supplementary methods specify covariance, backtracking and policy details.

Continuation retains the iterate, prior precision and computational prefix. At delivery time t_d, a candidate replaces the active version of its source and replays only acquired motion:

$$(x_{t_d}^{(j)}, P_{t_d}^{(j)}) = \mathcal{R}_{s \to t_d}(x_s^{(j)}, P_s^{(j)}; u_{s+1:t_d}). \tag{4}$$

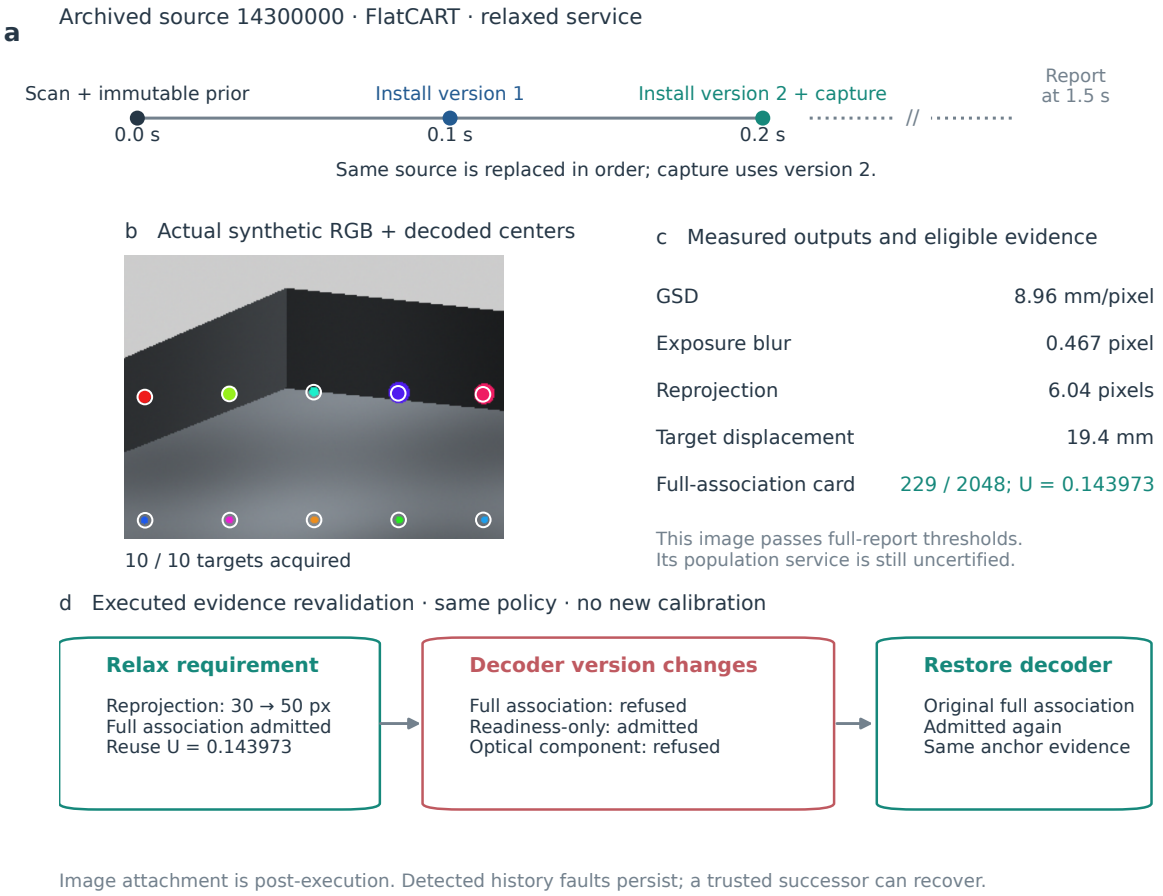


Figure 2: A complete archived source and executed software revalidation. (a) Actual numerical installations and capture use the frozen virtual clock. (b) The original synthetic RGB array with decoded-center overlays. (c) Measured outputs and the applicable full-association card. (d) Deterministic requirement/decoder fixtures reuse, withdraw and restore evidence with the policy held fixed. Panel (d) evaluates deterministic decoder-specification configurations against preloaded evidence.

Checkpoint revisions share one likelihood and prior. The receiver consequently replaces them in order; final rejection can restore the original source prior through the same interface. The serial service permits one active source job and drains pending publications before releasing it. Under a common retained solve and no interleaved independent update, delivery of the final version produces the same state as delivering that version alone at the same time. Figure 2 connects an archived numerical trace to the downstream task and its separately checked evidence.

3.3. Events, policies and computation budgets

For a four-second episode, integrated position loss is $L_\pi = \int_0^4 \|e_p^\pi(t)\|^2 dt$, evaluated by a 0.1-s right-endpoint sum. At study threshold $\epsilon = 0.10$ m, define

$$E_{\mathrm{tr}} = \{\max_{t\in\{0.5,0.6,\ldots,4.0\}} \|e_p(t)\| > \epsilon\}, \tag{5}$$

$$E_{\mathrm{obs}} = \{\text{no positive source version at } D\} \cup \{\|e_p(D)\| > \epsilon\}, \quad E_{\mathrm{un}} = E_{\mathrm{tr}} \cup E_{\mathrm{obs}}. \tag{6}$$

Same-tick ordered installation and rollback precede assessment. Trajectory, observation, both-marginal and union requests use these distinct events. A union request uses its directly evaluated OR event; two marginal

bounds yield only their summed union bound.

The retained CART and random-forest predictors share a 1,120-source development set, split into 926 training and 194 validation sources. CART has depth seven and minimum leaf size 20; the forest averages 64 depth-eight trees with minimum leaf size 12. Causal features describe uncertainty, information, residuals, steps, iteration, pair count, displacement and declared noise. Predictors estimate terminal MSE $\widehat{v}_{ja}$, remaining work $\widehat{c}_{ja}$ and a development-event frequency $\widehat{r}_{ja}$ for reachable cap a. The endpoint heuristic selects

$$\widehat{a}_j = \arg\min_{a \in \mathcal{A}_j} \left\{ T \max(\widehat{v}_{ja}, 0) + \lambda \max(\widehat{c}_{ja}, 0) + \omega \operatorname{clip}(\widehat{r}_{ja}, 0, 1) \right\}, \tag{7}$$

with $\lambda \in \{0.3, 1, 3\}$ m^2 and $\omega = 0.002$ m^{2}s. MSE comparisons also guide publication of eligible candidates. These forecasts are scheduling heuristics; event calibration concerns the complete executed wrapper.

The localization catalogue contains 22 policies: matched CART/RF variants at three prices, fixed caps, final convergence, residual and randomized controls, the earlier RAVC policy and time-only interventions. We retain the archived name KG1 for price-one contract-CART. FlatCART uses the same predictor; time-only and contract variants add deadline and profile predicates in sequence. Every policy receives the common next-stage budget reservation, acquisition and source-replay service. Hard-budget interruption is therefore part of each effective policy, including Fixed2.

The stage budget B covers source sealing, registration, features, forecasting, rules and publication. Total modeled work W additionally includes the common front end, receiver and stream. The frozen operation clock assigns CPU-equivalent costs to actual counted operations, then applies the declared load factor to delivery scheduling. Native CPU is a separate executed-work measurement. Selection uses the dimensionally consistent calibration objective $\overline{L} + \lambda_{\text{sel}} \overline{W}$, with $\lambda_{\text{sel}} = 1$ m^2. Supplementary methods give the installation-clock equation and the boundaries of component and whole-route CPU.

3.4. Calibration, admission and evidence lifetime

A typed contract records the source, numerical version, policy, population, service, requested event and their dependencies. Before opening the scan, a request fixes its event set and risk allowance δ. For a frozen event with k failures among n iid complete calibration inputs, the stored one-sided bound is

$$U(k, n) = \begin{cases} 1, & k = n, \\ \text{Beta}^{-1}(1 - \beta/K; k + 1, n - k), & k < n. \end{cases} \tag{8}$$

With $\beta = 0.05$ and localization family $K = 22 \times 2 \times 4 \times 3 = 528$, a union bound gives simultaneous risk coverage of at least $1 - \beta$. Dependence between policies or events on the same source is allowed. Calibration-dependent selection can use the resulting bounds because they hold for the entire frozen family on one confidence event.

An admitted policy matches the request's population, service and dependencies and satisfies $U \le \delta$ for every required event. The selector minimizes the same calibration loss–work objective and commits before sensing. An empty admissible catalogue refuses the statistical service. During execution, receipts check source sealing, ordered version identity, active dependencies and scope history. If $A_{s,j}$ is the current attachment candidate and $M_{s,j}$ the persistent history state,

$$M_{s,j+1} = M_{s,j} \wedge \text{validAttempt}_{s,j}, \qquad A^{\text{issued}}_{s,j} = A_{s,j} \wedge M_{s,j+1}. \tag{9}$$

Reconstructing a plan retains this source-scoped latch. Changed dependencies revoke affected cards; unrelated cards remain usable. The attachment observer modifies claim state while preserving canonical numerical decisions, installation ordering and modeled work. Its separately measured CPU and the noninterference proof are provided in the supplement.

The bound concerns unconditional events under the frozen program and stated deployment mixture. Applying it requires matched input distributions and trusted dependencies. Source checks establish execution identity; subgroup, receipt-conditioned and route risks require additional calibration. These conditions are used consistently throughout the evaluation.

Table 3: Reference contract procedure. Statistical commitment precedes sensing; ordered numerical execution and evidence attachment retain distinct state.

Step	Operation
1	Compile the task into required events, engineering units and dependency projections.
2	Match the population, service and effective policy against the applicable card inventory.
3	Validate each card and its containment witness; retain candidates whose required bounds satisfy $U \le \delta$.
4	Minimize $\overline{L} + \lambda_{\mathrm{sel}}\overline{W}$ and seal the selected policy and evidence before opening the source. Return `REFUSED` for an empty catalogue.
5	Execute the retained solve under the stage reservation; publish eligible versions in source order.
6	Replace the active source contribution, replay acquired motion, and check source/version identity, dependencies and persistent history.
7	Expose the task output and its applicable evidence status; withdraw affected statements after a detected invalid history or dependency change.

4. Changing inspection requirements

4.1. Acquisition, association and report outputs

The inspection extension retains the localization state, solver and replay interface. Its source-bound receipt observer checks frozen plans against archived execution records after numerical processing; its timing is reported separately. Up to 15 CAD face/barycentric addresses are planned before stochastic sensing. Raycasting creates RGB images with procedural texture, fixed lighting, 25-mm-radius diagnostic landmarks, a motion/exposure blur surrogate and Gaussian intensity noise. A nearest-palette decoder estimates landmark centers from pixels and known colors. Capture-time association uses the source version actually installed at the independently sampled capture tick $c \in \{0.1, 0.2, 0.3\}$ s.

At a planned target with range d, pixel focal length f_{px} and absolute incidence γ, geometric GSD is $g = d/(f_{\mathrm{px}}\gamma)$. We score the maximum GSD and blur over fixed planned addresses, acquired coverage from decoded landmarks, reprojection error in pixels, CAD-target displacement and positive-version readiness. At speed v and exposure τ, the target blur proxy is $\max_i v\tau/g_i$; the raster uses a Gaussian exposure surrogate characterized by its pixel-median GSD. For nominal CAD target q_i and its physical counterpart $q_i^\star$, the metre-valued association quantity is

$$a_{\mathrm{assoc}} = \max_{i \in \mathcal{I}_c} \left\| \widehat{m}_R q_i + \widehat{m}_t - (m_R^\star q_i^\star + m_t^\star) \right\|, \tag{10}$$

where $\mathcal{I}_c$ contains visible in-ROI targets with finite decoded centers and positive predicted depth. Reprojection separately compares pixels predicted from the active joint state with decoded centers. These scored quantities retain distinct units and dependencies. Full acquisition requires GSD at most 20 mm/pixel, blur at most 0.5 pixel and coverage at least 0.50. Full association requires coverage at least 0.50, reprojection at most 30 pixels, displacement at most 0.10 m and readiness at capture.

Full reporting adds delay at most 2 s and image age at most 1.8 s. With the frozen virtual raster/decoder work w_{camera} and load α,

$$t_{\mathrm{report}} = \max\{t_{\mathrm{first\ positive}}, c + 0.05 + \alpha w_{\mathrm{camera}}\}, \qquad a_{\mathrm{image}} = t_{\mathrm{report}} - c. \tag{11}$$

This clock characterizes the virtual producer. The declared thresholds define synthetic diagnostic services. Their complete eight-variant specification and all unit checks are supplied in the supplement.

4.2. Requirement containment and producer dependencies

A requirement R compiles into admissible intervals and a failure event:

$$S_R = \{z : \ell_{R,j} \le z_j \le u_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{12}$$

Nonfinite required outputs fail; the compiler rejects invalid specifications and incompatible units. An anchor A covers R when $J_R \subseteq J_A$ and $\ell_{A,j} \geq \ell_{R,j}$, $u_{A,j} \leq u_{R,j}$ for each requested quantity. The resulting witness establishes

$$S_A \subseteq S_R \quad \Longrightarrow \quad E_{p,R} \subseteq E_{p,A} \quad \Longrightarrow \quad \Pr(E_{p,R}) \leq U_{p,A}. \tag{13}$$

On the simultaneous anchor-coverage event, this implication holds for the entire containment closure. Calibration-dependent anchor selection within that closure adds no separate event family. Dropping a required quantity defines a partial service and is recorded explicitly.

Producer changes require an additional condition. Let D_j be the frozen implemented dependency set of quantity j, and $D_R = \bigcup_{j \in J_R} D_j$. An event capsule identifies the component snapshot projected onto D_R. Transfer requires the requested functions to remain identical on supported complete inputs:

$$z_{p',j}(X) = z_{p,j}(X) \quad (j \in J_R) \quad \Longrightarrow \quad E_{p',R} = E_{p,R}. \tag{14}$$

Together with containment and an unchanged population, this equality preserves the old anchor bound. Producer compatibility is evaluated against the trusted implemented dependency specification. The three rules serve distinct implementation decisions: containment authorizes a changed requirement, functional invariance retains evidence after an unrelated provider change, and source-history validity permits attachment. The following theorem composes these rules into the correctness property of the operating mechanism.

Theorem 1 (Coverage of an authorized event)**.** *For one frozen calibration family, consider a request committed before sensing. Let a valid anchor A cover its requested failure event E_R, with $U_A \leq \delta$. Assume the deployment population is matched and the requested producer functions remain invariant. On the simultaneous calibration-confidence event, $\Pr(E_R) \leq \delta$. For an issuance indicator I generated by the source-history monitor,*

$$\Pr(E_R \cap \{I = 1\}) \leq \delta. \tag{15}$$

A detected invalid attempt makes issuance false for subsequent attachments in that source scope while preserving valid numerical installation.

Proof. Requested-function invariance preserves the request event, and the containment witness gives $E_R \subseteq E_A$. Simultaneous anchor coverage yields $\Pr(E_R) \leq \Pr(E_A) \leq U_A \leq \delta$. Intersecting with the issuance event preserves the upper bound. The conjunction in Eq. (9) remains false after a failed attempt, independently of later numerical installations. □

The simultaneous confidence event has calibration probability at least $1 - \beta$. Equation (15) bounds the joint event of task failure and evidence issuance. The theorem justifies the implemented reuse and issuance decisions through the existing finite-family bound; its state property ensures that plan reconstruction preserves a detected invalid history.

A wider ROI changes acquired coverage, association outputs and camera work. At the same planned addresses, fixed pixel focal length and target geometry preserve geometric GSD. Optical evidence can therefore remain applicable while a complete image or report event requires renewal. Capture-time readiness similarly uses the version available at capture; future pose updates do not retroactively alter it. Policy and anchor commitment before sensing avoids per-image switching among marginally calibrated events.

4.3. Fair evidence-reuse alternatives

Contract coverage uses explicit containment witnesses and dependency capsules. Complete procedural coverage independently implements the same interval and dependency semantics. Cached direct renewal evaluates predeclared requested events using applicable calibration records, including invariant pools available to the other methods. Exact-card matching is the limited anchor-identity control. All receive the same quantities, cards, confidence allocation, records, objective and request cache. We evaluate cold preparation separately from first-query validation and warm lookup. Their archived method identifiers and complete specifications are supplied in the supplement.

Table 4: Shared processing services and distinct virtual camera work. Stage budgets concern numerical sender work; total work includes additional common components.

Service	Load α	Deadline D (s)	Stage budget B (ms)
Relaxed	10	0.5	20
Standard	20	0.3	12
Tight	40	0.2	6
Contended	60	0.3	12
Virtual raster + decoder: A 62.5 ms; B 125 ms			

5. Experimental design

5.1. Independent inputs and prospective freezes

The localization study contains 8,192 calibration inputs, 4,096 independent confirmation inputs and 1,536 unsupported stress inputs. Nominal geometry mixes B767 and C172; the separate unfitted airframe is excluded from predictor training and receives matched fresh calibration. Each profile has 4,096 calibration and 2,048 confirmation sources. Forty-eight pilot sources are excluded. Numerical policies, population definitions, events and the operation clock were frozen before formal generation; cards and requests were sealed before confirmation. A disclosed pre-confirmation source-history amendment preserves the numerical/clock/event projection and its versioned audits.

The two inspection studies use independent sources and retain separate evidence. Study A renders 192×144 pixels and B renders 288×216, both with focal length 1,800 pixels and exposure 6 ms. Each contains 4,096 calibration, 2,048 confirmation and 256 stress sources, totaling 12,800 across A and B. Invariant optical events can pool the independently identified calibration blocks; other events use the matched study. The reserved family is $K = 4,608$ with $\beta = 0.05$, of which 3,072 cards are stored. Unused reserve remains allocated. All 6,144 plans are sealed before confirmation.

Policy, service, event and request repetitions are paired executions of these sources. They add no independent sample size. Nominal mixtures sample prior deviations of 0.015/0.03/0.06 m, noise multipliers 1/2 and shape levels 0/1/2. Separate unsupported stress introduces occlusion, range degeneracy, larger prior error or map warp. The supplement records seeds, perturbations, freezes, amendments and full execution lineage.

5.2. Comparisons and measurement boundaries

Matched module interventions separate predictor decisions, time/profile predicates and attachment checks. Full localization selection is compared with a task-aware calibrated procedure using identical source/version checks. The primary 16 paired intervals cover two profiles, four contrasts and two metrics at standard service, using 32,000 source-bootstrap resamples. Pointwise and Bonferroni-adjusted percentile intervals are approximate sampling intervals; exact binomial bounds govern admission. Sensitivity, subgroups, Pareto means and stress remain descriptive.

The route study executes 36 distinct 600-s inputs: three geometries, three conditions and four realizations, with 12 methods and 299 scan opportunities per route. It measures the complete live online loop, with source generation, archive hydration and truth scoring outside the timer. Inspection component measurements and preloaded-record preparation have different boundaries. The Windows camera CPU observations have 15.625-ms quantization; modeled work, component CPU and complete-route CPU use their stated measurement boundaries.

A supplementary deterministic study holds the precommitted policy fixed in all 320 unique primary A/B contexts and applies 16 configurations per context: unchanged, 13 individual provider edits and two paired edits. Contract coverage, complete procedural coverage and cached direct renewal use identical applicable evidence. An independently coded interval/dependency oracle checks withdrawal and restoration under the same declared dependency model. These 5,120 paired configurations yield 15,360 implementation decisions,

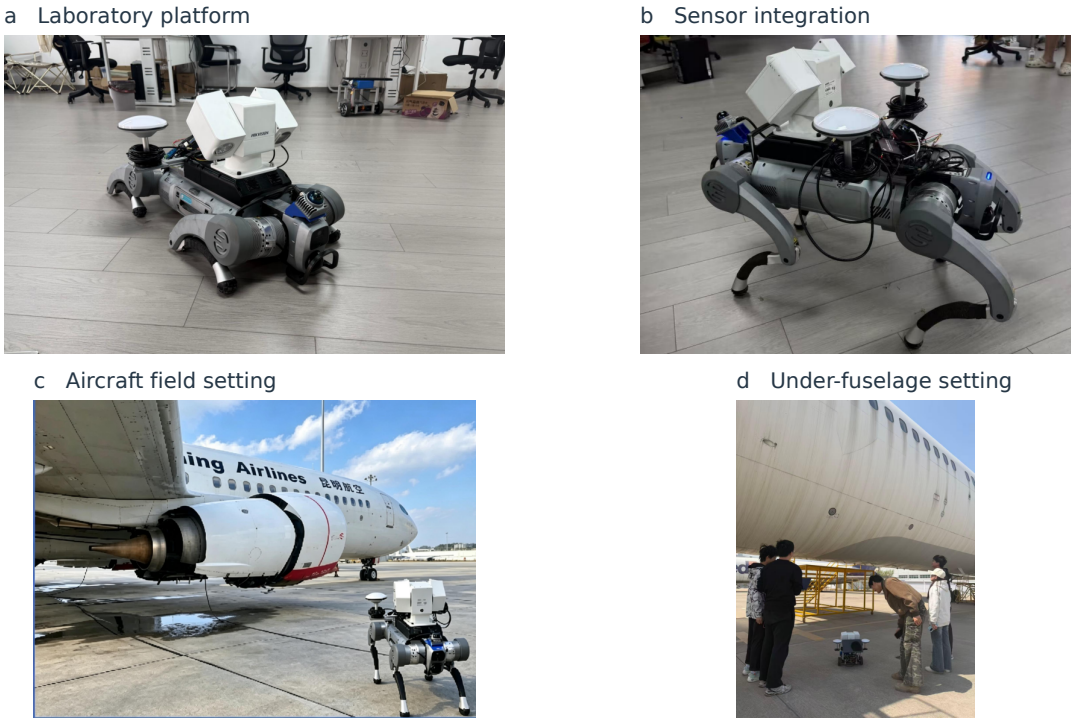


Figure 3: Original platform photographs: laboratory integration, sensor integration, aircraft field setting and under-fuselage setting. These photographs document engineering context; quantitative results in this study use the declared synthetic inputs. Original image proportions are retained.

with no new population inputs or calibrated cards. The protocol, source hashes and disclosed illustrative refinement are preserved in the supplement.

Figure 3 preserves the four original platform photographs as engineering context. Synthetic raster landmarks supply the evaluated image task. Complete current-study tables, original simulation views, routes, tail summaries and fault fixtures are retained in the supplement and reproducibility assets.

6. Results

6.1. Service support under reuse and direct renewal

At allowance 0.15, anchor reuse supports 24/192 configured labels in A and 26/192 in B; direct renewal supports 30/192 and 32/192. Several variants compile to the same event. Counting unique (population, service, event) contexts gives 24/160 and 23/160 for reuse, versus 30/160 and 29/160 for renewal (Table 5). The label and unique-event counts therefore describe complementary aspects of service support. Figure 4 retains every label and the additional direct-renewal services.

The full-association primary plan for B/unfitted/relaxed uses FlatCART. It has calibration count $229/2,048$, upper bound 0.143973 and independent confirmation count $103/1,024$. The admission margin is 0.006027. Full acquisition and reporting remain unsupported; other accepted requests are explicitly partial or component services.

6.2. Evidence-preparation work on matched requests

Matched accounting holds support and policy fixed on all 47 unique contexts admitted by reuse. The selected anchor catalogue needs 49,152 event–record evaluations and 23 proof cards, versus 61,440 and 29 for

Table 5: Primary-allowance service support. Configured labels retain the frozen request catalogue; unique contexts deduplicate compiled event identities within population and service. Full tasks retain all original requirements.

Study	Method	Labels /192	Unique /160	Full /24
A	Contract coverage	24	24	0
A	Complete procedure	24	24	0
A	Cached direct renewal	30	30	0
A	Exact-card control	18	18	0
B	Contract coverage	26	23	1
B	Complete procedure	26	23	1
B	Cached direct renewal	32	29	1
B	Exact-card control	20	17	1

Table 6: Executed preloaded-record preparation and query costs. Cold preparation is the median of three trials; the first batch has 384 queries. Warm time is per shared-cache lookup.

Method	Cards	Cold (s)	First batch (s)	Warm (μs)
Contract	1,152	0.179	6.408	0.091
Complete procedure	1,152	0.176	6.225	0.111
Cached direct renewal	2,560	0.365	9.991	0.098
Exact-card control	1,152	0.186	8.747	0.103

direct event preparation: a 20% reduction (Fig. 5a). This comparison isolates evidence-preparation work for the same services. Across the complete catalogues, anchor preparation evaluates 2,490,368 records for 1,152 unique cards, versus 5,373,952 records and 2,560 cards for direct renewal: 53.66% fewer event–record evaluations, with narrower request support (Fig. 5b). These are counted catalogue-construction requirements; warmed caches require no rescoring. Table 6 retains actual preloaded-record timings. All methods then receive the same warm request cache, so recurring dictionary lookups avoid record rescoring. Seventy-two producer-edit fixtures and 32 malformed-receipt fixtures verify specified invalidation and persistent refusal; fresh trusted execution scopes remain admissible.

All 3,072 task cards reproduce their exact binomial bounds with the original reserved-family allocation. Figure 5c shows fixed-event sensitivity under nested prefixes. Passing a full-sample association card leaves a narrow margin; smaller prefixes often fail admission. These diagnostic prefixes retain the frozen plans.

6.3. Concrete source, dependency change and service recovery

The archived B/unfitted source 14,300,000 supplies a complete numerical and image case (Fig. 2). FlatCART installs versions one and two at 0.1 and 0.2 s; capture at 0.2 s consumes version two. All ten planned targets are acquired, reprojection is 6.04 pixels and CAD-target displacement is 19.4 mm. A prespecified consumer relaxation from 30 to 50 pixels remains covered by the original full-association anchor, reusing $U = 0.143973$ without recalibration.

Changing the decoder specification withdraws full association. A readiness-only request retains an applicable component bound $U = 0.056804$, whereas the optical component remains refused under its archived risk bound. Restoring the original decoder restores full association. This source individually passes the full-report thresholds, while the population-level full-report service remains uncertified: a favorable image does not authorize a missing population statement.

Figure 6 resolves change decisions by edited provider. The contract matrix contains 47 initially admitted contexts. Across their 705 producer edits, 489 require withdrawal and 216 retain applicable evidence; every edit and restoration agrees with the declared evidence rules. Unchanged controls and initially refused requests remain in the 5,120-configuration matrix per method. Complete results for all three implementations are retained in the supplement. Diagnostic stale-plan reuse would incorrectly retain 489 contract attachments; whole-producer withdrawal would unnecessarily refuse 216. In 24 additional packet-fault cases, refusal

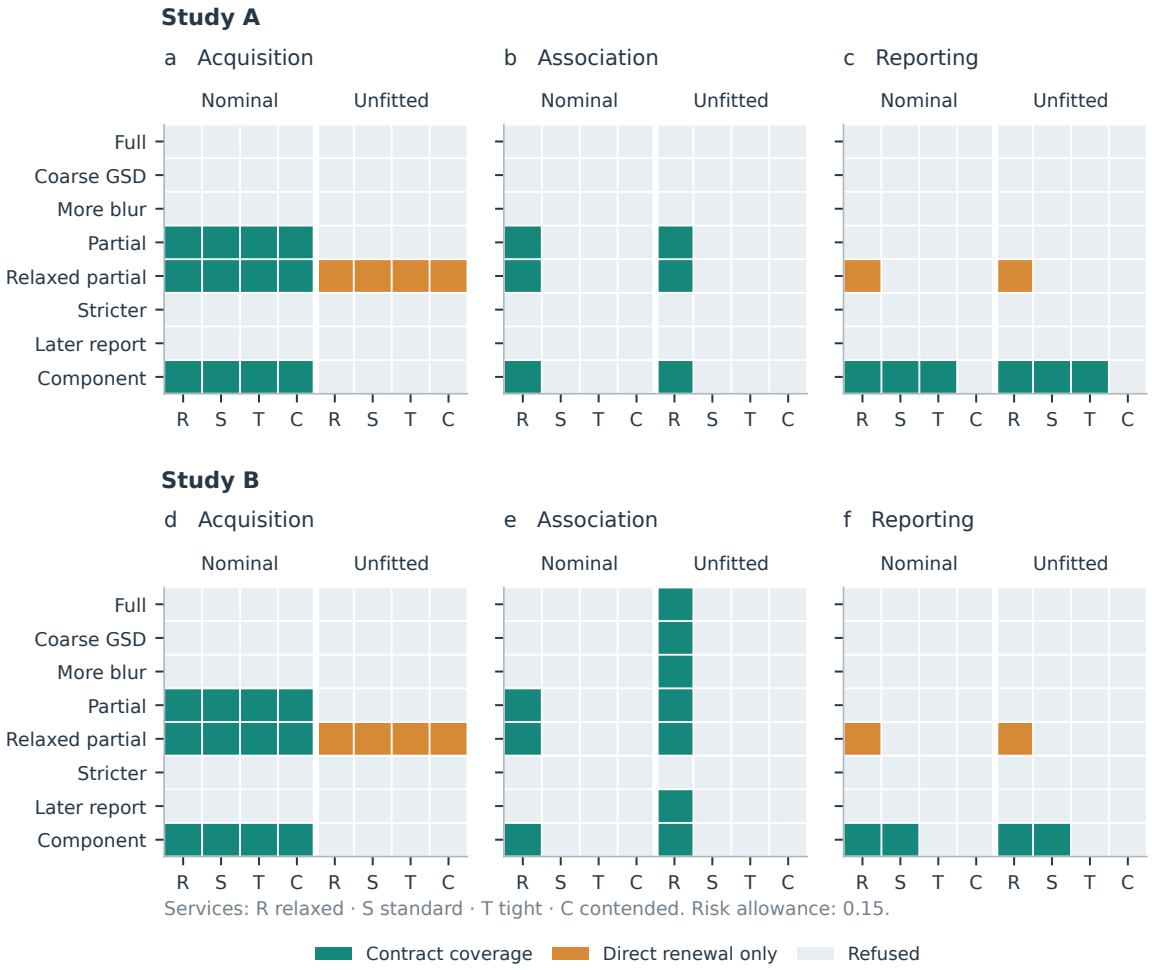


Figure 4: All 192 primary request labels per study. Teal cells show contract coverage; orange cells show additional cached direct-renewal support; grey cells are refused. Repeated event identities remain in this configured-label display and are separately deduplicated in Table 5.

survives gate reconstruction and a fresh trusted execution scope restores attachment. These deterministic configurations test the specified source-history protocol.

6.4. ROI changes separate geometric quality from delivery feasibility

The excluded paired pilot in Fig. 7 uses the same source and landmarks for both ROIs. Across 48 paired pilots, fixed-target GSD is invariant: wider ROI changes acquired coverage and virtual production work while range, incidence and focal length remain fixed. The formal A/B confirmation inputs are independent; their empirical differences are not a paired causal estimate. Optical projection invariance permits selective reuse, while affected coverage, association and report events require their own evidence.

The reporting clock also creates an analytical infeasible region. Equation (11) gives

$$a_{\text{image}} \geq 0.05 + \alpha w_{\text{camera}}. \tag{16}$$

This lower bound exceeds the full-task 1.8-s limit for A tight/contended and B standard/tight/contended: five of eight study/service combinations (Fig. 5d). Their reporting failure is built into the declared timing

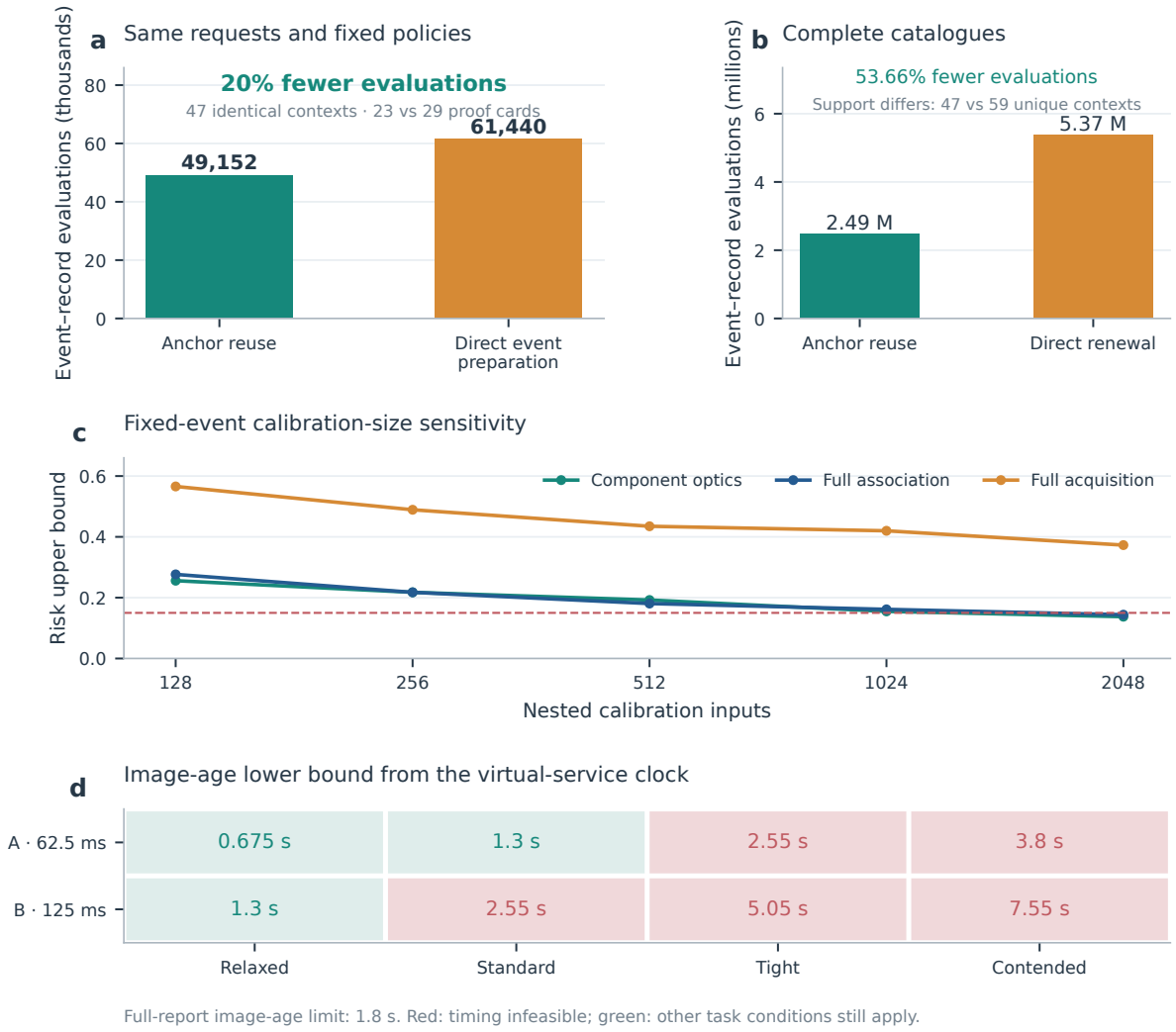


Figure 5: Evidence-preparation work and service feasibility. (a) The same 47 supported contexts and fixed policies require 49,152 event–record evaluations under anchor reuse versus 61,440 under direct event preparation, a 20% reduction. (b) Complete catalogues have different support and show 53.66% fewer evaluations under reuse. (c) Fixed-event bounds under nested calibration prefixes. (d) Exact image-age lower bounds; five service combinations exceed the 1.8-s full-report limit.

model. The other three combinations permit timing but can fail acquisition, association or actual delivery conditions. All full-report requests remain refused at the primary allowance.

6.5. Localization accuracy, work and availability

At standard service, KG1 reduces integrated loss relative to Fixed2 by 15.68% on nominal geometry and 12.61% on unfitted geometry, with work increases of 16.09% and 16.07% (Table 7). Fixed4 and Final further reduce mean loss at greater work, while the RF policy remains close to Fixed2. Figure 8 displays all 22-policy means and the prespecified paired intervals, covering the four matched method contrasts.

At the primary allowance, Full admits 18 of 32 localization request contexts. Relaxed and standard services admit all four request types. Contention admits trajectory-only requests; tight service admits none.

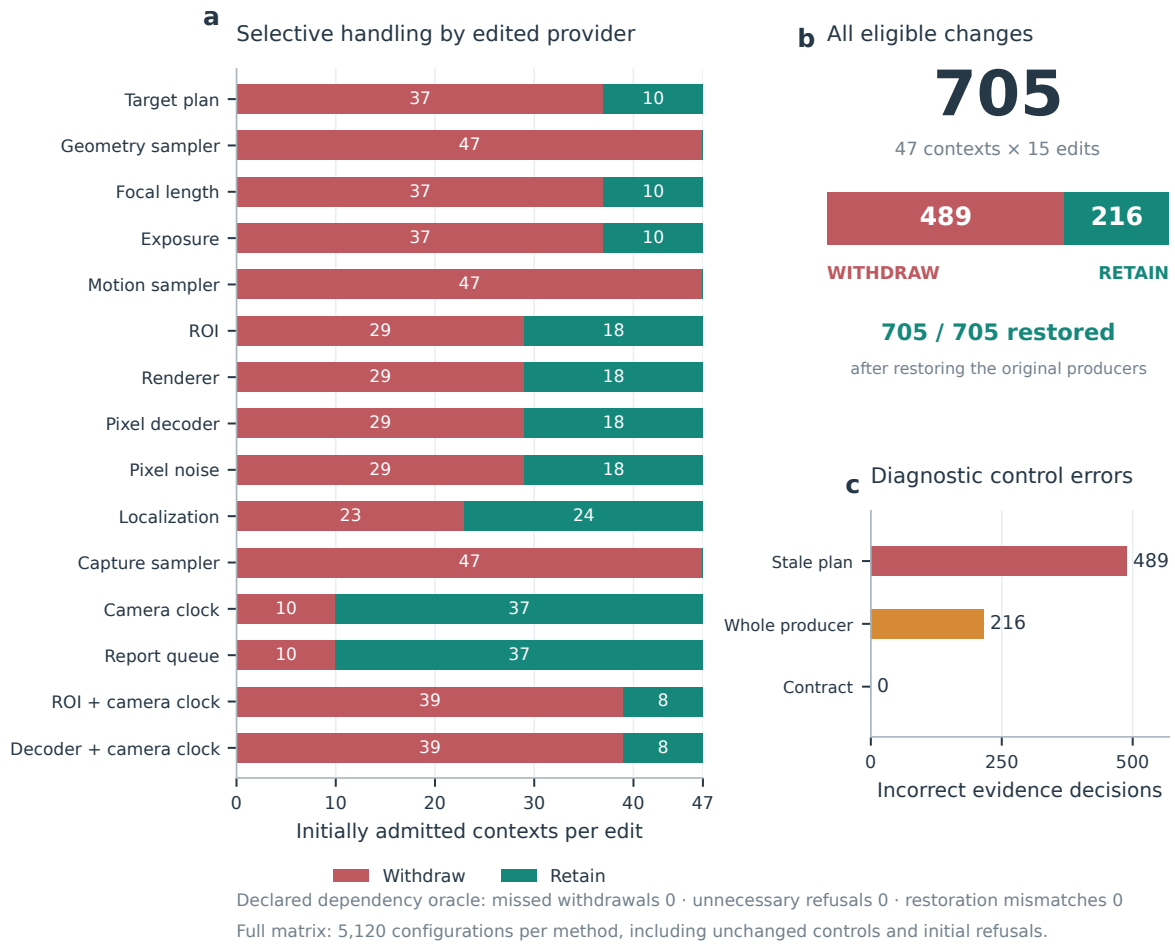


Figure 6: Evidence decisions after producer changes. (a) Each of 15 edit configurations is applied to the same 47 initially admitted contexts; bars show requested-dependency withdrawal and retention. (b) The 705 edited contexts contain 489 withdrawals and 216 retained statements. (c) Diagnostic stale-plan reuse misses 489 withdrawals, while whole-producer withdrawal unnecessarily refuses 216; the contract decisions agree with the declared dependency oracle. Original producers restore all 705 initially admitted edit contexts. Unchanged controls and initial refusals remain in the full 5,120-configuration matrix per method.

Refused requests remain in support denominators and retain the propagated prior during the corresponding execution.

The gains also depend on episode difficulty. The two largest nominal contributions account for 27.99% of the KG1–Fixed2 net loss gain; the unfitted share is 10.02%. Diagnostic removal leaves positive mean gains of 0.001563 and 0.002265 m^2s, while the primary comparison retains every source. Complete tail and unsupported-stress results are provided alongside these means.

6.6. Action predicates and evidence checks

Time screening changes every CART stopping endpoint under contention. On nominal confirmation it reduces work by 3.600 ms and observation failure by 1.660 percentage points while increasing loss by 0.005094 m^2s. The unfitted changes are −3.583 ms, −1.904 percentage points and +0.004179 m^2s. The common next-stage reservation limits refinement under the tight budget. The supplement plots all-service time-predicate effects and the original attachment/dependency fixtures.

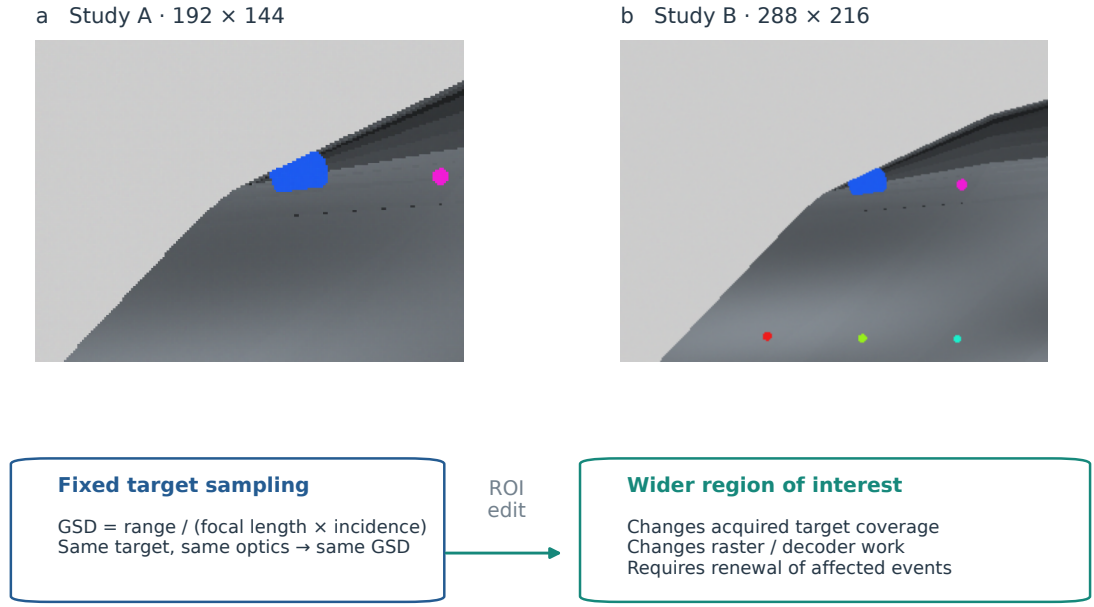


Figure 7: Actual archived synthetic raster output from excluded pilot 12000000 under the paired ROI intervention, with diagnostic landmarks and source fixed. The original displayed rasters are reproduced from their archived figure assets. Wider ROI changes coverage and work; fixed-target GSD follows the unchanged optical/geometry projection.

Table 7: Standard-service confirmation, $n = 2,048$ per population. L is integrated loss in 10^{-3} m^{2}s; W is total modeled work.

	Nominal			Unfitted		
Policy	L	W (ms)	Union fail (%)	L	W (ms)	Union fail (%)
Fixed2	13.833	9.221	8.64	19.938	9.220	10.06
Fixed4	11.295	12.109	7.52	17.350	11.736	9.28
Final	11.295	12.109	7.52	17.350	11.736	9.28
FlatCART	11.664	10.705	6.84	17.423	10.702	8.79
KG1	11.664	10.705	6.84	17.423	10.702	8.79
FlatRF	13.844	9.973	8.64	19.936	9.953	10.06
ContractRF	13.844	9.973	8.64	19.936	9.953	10.06
OldRAVC	12.199	9.666	6.79	17.964	9.781	9.03
Full	12.301	9.721	7.13	17.964	9.781	9.03
Complete procedure	12.301	9.721	7.13	17.964	9.781	9.03

Evidence checks affect attachment correctness. Across 16 constructed invalid cases using actual frozen cards, stripped checks issue 16 incorrect attachments. Matching reduces this to four, current-version validation to two, and persistent source history to zero. The valid controls retain their attachments. Dependency edits selectively revoke cards: changing the RF forecast affects 168 of 528, whereas a shared clock change affects all 528. These fixed software fixtures measure compliance with the specified protocol.

6.7. Continuous-route accuracy and availability

Figure 9 shows all 12-method route summaries and highlights Fixed2, KG1 and Full. Ordinary routes often favor Fixed2 on CPU or accuracy; modest refinement opportunities provide limited return on forecasting

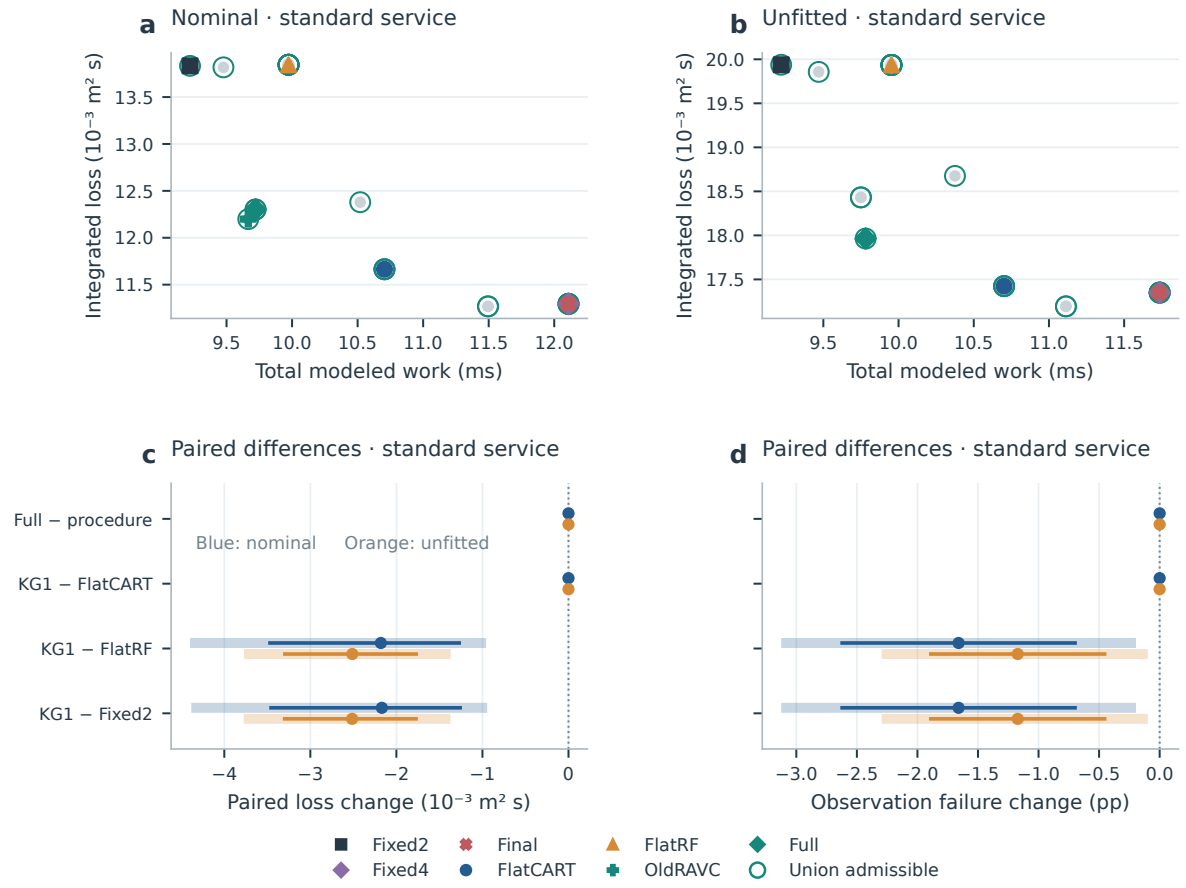


Figure 8: Standard-service localization. (a,b) All 22 policy means, highlighted controls and full selection; green rings mark direct-union eligibility at allowance 0.15. (c,d) All 16 paired intervals, colored by population: thin intervals are pointwise 95%, pale intervals are family-adjusted percentile intervals.

and repeated publication. Under C172 contention, KG1 raises timely-correction success from 53.01% to 80.94%, with RMSE increasing from 5.237 to 5.341 cm. Full achieves 81.61% and 5.234 cm. Whole-online CPU is 2.121, 2.194 and 2.028 s, respectively, over each 600-s route.

Severe stress produces large route errors across the catalogue. The route selector refuses episode-to-route attachments and continues an explicit provisional recommendation. All 432 executions remain descriptive route evidence; their samples and policy switches are separate from episode calibration. Full tables and the predeclared continuous trace are supplied in the supplement.

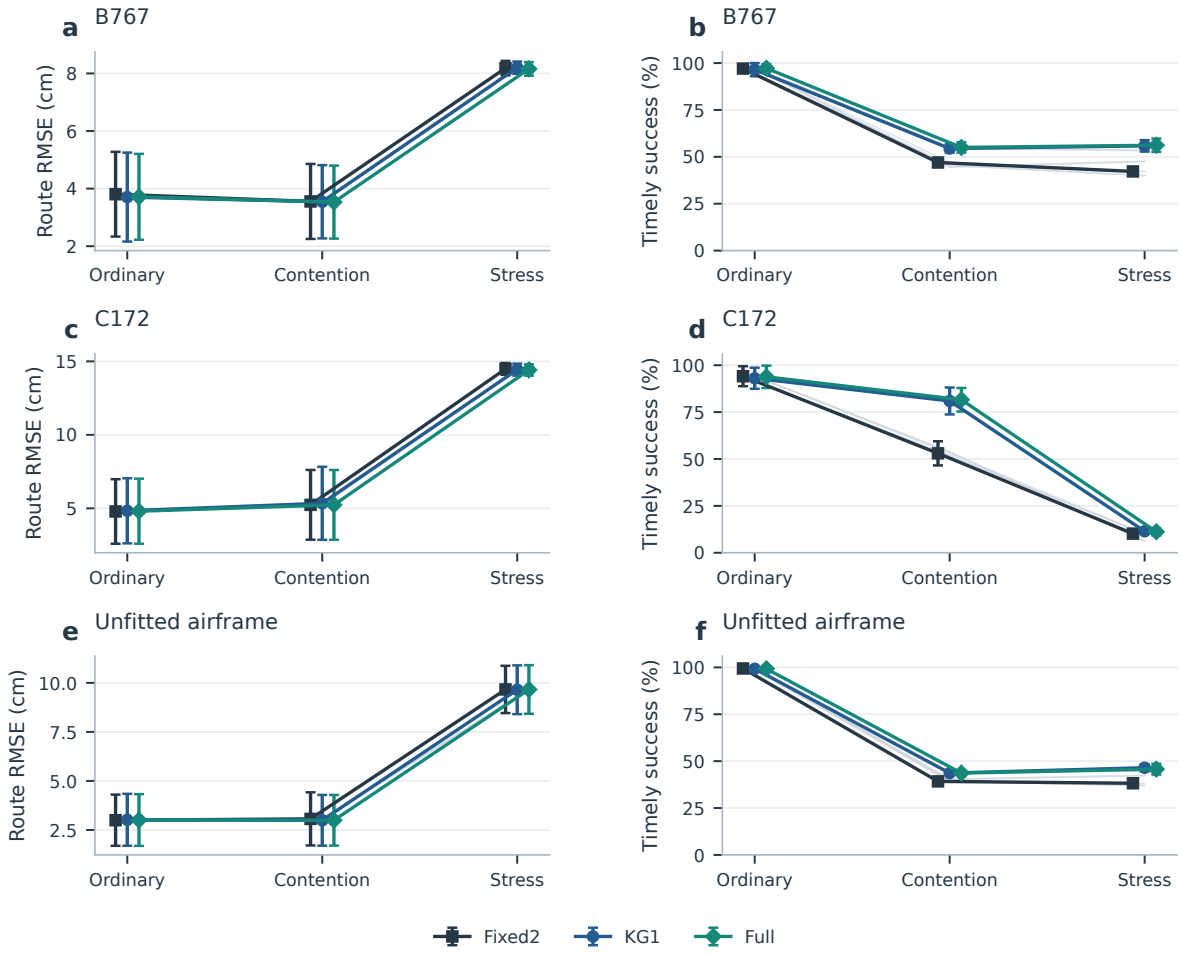


Figure 9: Complete-route results by geometry and condition. Grey traces retain all 12 methods; colored traces highlight Fixed2, KG1 and Full. Each point summarizes four matched 600-s inputs, with sample-standard-deviation bars. Route summaries carry no transferred episode bound.

7. Discussion

The operating contract makes statistical evidence an explicit engineering object with a task, numerical source and producer lifetime. A containment witness explains why a relaxed requirement can reuse its anchor. A dependency projection explains why a changed decoder withdraws association while a supported readiness component remains available. The source-history latch explains why reconstructing a plan cannot erase an invalid attachment attempt. Together these operations turn change information into inspectable service decisions.

The evaluation separates the value of reuse from its support–cost trade-off. On identical supported contexts and fixed policies, the selected anchor catalogue reduces event–record evaluations by 20%. The broader catalogues show a larger preparation reduction alongside fewer supported events. Anchor reuse and direct renewal can therefore coexist: reuse serves covered requests from existing proofs, while renewal supplies directly calibrated events outside that closure. In the change matrix, selective handling retains 216 applicable statements while withdrawing 489 affected ones, avoiding both stale evidence and unnecessary whole-producer refusal.

Numerical execution supplies the interface quantities. Ordered replacement preserves one likelihood

per source, and capture binds association to its installed version. Fixed policies remain economical when refinement has little return; time screening improves availability under contention at a measured loss cost. These trade-offs determine the numerical service catalogue.

The representation also guides engineering changes. Wider ROI affects coverage and camera work while preserving fixed-target GSD. The image-age boundary identifies timing-infeasible report services. The task and dependency specification thus distinguishes a change requiring renewed event evidence from one requiring a faster producer.

8. Conclusion

Executable operating contracts connect changing inspection requirements to the applicability and lifetime of statistical evidence. The typed representation binds task events, committed policies, numerical source versions, producer dependencies and calibrated cards. Containment, invariant requested functions and persistent source history establish the mechanism's correctness conditions for reuse and issuance. Evaluation shows a 20% reduction in evidence-preparation event–record evaluations on the same 47 supported contexts and fixed policies, with 489 selective withdrawals and 216 retained statements across 705 producer edits. Independent localization and image studies supply the underlying accuracy, work, availability and service-support results. The resulting framework provides an executable, traceable evidence lifecycle for aircraft localization and inspection.

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Data and code availability

The reproducibility release retains protocols, independent source seeds, native solver and predictors, execution records, risk cards, amendment history, full result tables and editable source. The manuscript package includes table snapshots, the deterministic change/recovery validator, SHA-256 provenance and reproducible PDF/SVG scripts. Historical source data and original photographs remain separate from the AI-assisted conceptual illustration.

Declaration of generative AI and AI-assisted technologies

OpenAI ChatGPT/Codex assisted code development, computational analysis, manuscript revision and exact data-figure scripts. OpenAI image generation produced the conceptual illustration in Fig. 1a from a platform-shape reference; it contributes no experimental observation. The concept panel is explicitly labeled; quantitative graphics use archived numerical observations.

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Evidence Boundary

Baseline equivalence and attribution. The standard-service numerical gains of KG1 are shared by FlatCART. Fixed4 and Final also have coincident means under the common budget wrapper, and the FlatRF/ContractRF rows retain their observed values. Across relaxed and standard confirmation, FlatCART, time-only CART and KG1 have identical numerical outcomes; the profile predicate changes none of the studied confirmation outputs. The time predicate changes decisions under contention, as reported in the mechanism intervention.

The complete procedural implementation reproduces Full's numerical and receipt outcomes, the 18/32 primary localization admissions, and the inspected continuous-route numerical trajectories. In the 16 invalid attachment fixtures, both persistent-history implementations have zero incorrect attachments and accept the valid controls. Contract coverage and complete procedural coverage agree on all 6,144 frozen image-task plans and repeated confirmation outputs, including the 24/23 supported unique primary contexts. Their change/recovery decisions also agree. The complete procedure shares the anchor-preparation reduction. These controls attribute numerical gains to scheduling and evidence-management gains to the implemented semantics; they establish an executable reference model rather than a separate numerical advantage from the contract representation. The tables, paired intervals, request maps and diagnostic traces retain the corresponding coincident results.

Simulation evidence and physical implementation. The quantitative evidence consists of the declared synthetic localization, route and image-task inputs and deterministic software configurations. Original platform photographs provide engineering context. The AI-assisted conceptual illustration contributes no experimental observation. The virtual operation clock, Windows component CPU and whole-route timing retain their stated measurement boundaries. Physical implementation requires representative observations, platform-specific service timing and matched population calibration. Comparative human maintenance effort and physical fault rates require their corresponding engineering studies. These activities belong to implementation and validation of the reference framework.

Scope of evidence reuse and runtime monitoring. Bounds describe unconditional events under the frozen effective program and supported population. Their application assumes iid complete calibration inputs, a matched deployment distribution and trusted implemented dependencies. Receipt-conditioned, geometry-subgroup and continuous-route risk statements require their own calibration. The joint issuance bound in Theorem 1 is distinct from a risk bound conditional on a received statement. A finite paired fixture checks the declared change; functional invariance is the premise for transfer to a producer with an unchanged requested projection.

Image-task attachment is evaluated after numerical execution. Decoder-version edits in the change/recovery study revalidate the component specification and existing evidence; they introduce no newly rendered population experiment or calibrated cards. Packet-fault recovery tests create a fresh trusted execution scope with the archived source seal, so they establish scope-level recovery. The monitor handles the specified detected history failures. Hidden distribution changes, undeclared physical couplings and compromised trusted code require additional mechanisms. The full-report service remains refused at the primary allowance, including cases where an individual image meets its thresholds. Improving that complete service requires a producer intervention and calibration under the stated full requirements.